

ACTIVITY 20 — FACILITATOR GUIDE**Earned Value Management and Schedule Compression**

TYPE	DURATION	MAPS TO	TOPIC
Workshop	60 minutes	LO5	5 - Monitor and Control the Project

Purpose of this activity

By the end of this activity learners will be able to compute every core EVM measure from the real Contoso cost baseline: CV, SV, CPI, SPI.

What learners produce

evm calculations, forecast, compression decision — submitted to artifacts/.

Materials required

- Learner Worksheet (one per learner)
- Checklist (one per group)
- Flipchart or A3 sheet and markers
- Laptop with internet access for the live problem-solving tools
- The course slides for the framework and worked example

Set-up

Groups of 3 to 4. Distribute one worksheet per learner and one checklist per group; each group needs one laptop for the calculations.

Scenario / briefing material

Contoso Learning Pte Ltd is a training provider with 14,000 active learners. Its course registration portal is 9 years old. Registration takes a median of 11 minutes and 34% of attempts are abandoned. Learner communications consume 24 staff-hours per week across 3 administrators. 62% of learner traffic is mobile, against a portal that is not responsive. Legacy consent screens do not meet current PDPA requirements. The project modernises course registration and learner communications.

How to run it — step by step

#	Instruction to learners	Facilitator note
1	Take the Contoso baseline and actuals. This is the real time-phased cost baseline from Lab 13, together with the measured EV and AC at each fortnightly reporting point. The project is at the end of week 14, which is the end of sprint 7 of 12.	Check the arithmetic on one group's numbers before moving on — a wrong sign propagates into every later step.
2	Compute the variances and indices at week 14. Work these by hand before checking. Use the cumulative figures - EVM is always cumulative unless a question explicitly says "this period".	Make a group state the UNITS and the base date. Most errors here are unit errors, not maths errors.
3	Forecast the outcome. Compute all three EAC variants, then choose the defensible one.	Ask what the number would have to be before the decision flips. That is the exam's favourite

		question.
4	Analyse the trend, not the snapshot. A single period tells you little. Compute CPI and SPI for every period and look at the direction.	Check the arithmetic on one group's numbers before moving on — a wrong sign propagates into every later step.
5	Check the escalation threshold. Go back to your Lab 03 escalation thresholds and apply them. This is the step that converts analysis into required action.	Make a group state the UNITS and the base date. Most errors here are unit errors, not maths errors.
6	Decide: crash or fast-track. The Board asks how the schedule position (SPI 0.929) will be recovered. You have two levers, and they fail in different ways.	Ask what the number would have to be before the decision flips. That is the exam's favourite question.
7	Write the variance report. Use the Lab 18 sponsor format. State the position, the cause, the action and the ask.	Check the arithmetic on one group's numbers before moving on — a wrong sign propagates into every later step.
8	Answer the exam-style scenarios	Make a group state the UNITS and the base date. Most errors here are unit errors, not maths errors.

Suggested timing

Minutes	Phase
6 min	Brief the scenario, confirm the inputs and assign roles
22 min	Work the first half of the steps individually, then pool as a group
23 min	Complete the remaining steps and finalise the deliverable
9 min	Group share-back, facilitator debrief and exam-style linkage

Total: 60 minutes

Discussion & decision prompts

1. Compute every core EVM measure from the real Contoso cost baseline: CV, SV, CPI, SPI — how did your group decide, and what would change that decision?
2. Forecast completion four different ways - EAC, ETC, VAC, TCPI - and know which EAC formula applies to which situation — how did your group decide, and what would change that decision?
3. Read the CPI/SPI trend over seven periods rather than judging from a single snapshot — how did your group decide, and what would change that decision?
4. Your cumulative PV at period 12 equals SGD 480,000 exactly. If it does not, stop - every forecast downstream is invalid — can every member of your group defend it?
5. Where would earned value management and schedule compression have saved you time or money on a real project you have worked on?

What good looks like

- Your cumulative PV at period 12 equals SGD 480,000 exactly. If it does not, stop - every forecast downstream is invalid.
- $CV = -22,000$ and $SV = -19,000$, both negative, and you described SV in dollars rather than days.
- $CPI = 0.919$ and $SPI = 0.929$, and you carried CPI to at least four decimal places before computing EAC.
- You computed all three EAC variants and *chose* one with a stated reason drawn from the trend, rather than defaulting to the first formula you remembered.
- Your TCPI of 1.105 is compared against your CPI of 0.919, and you concluded in writing that recovery to the original budget is not credible.
- You identified the seven-period monotonic CPI decline as systemic, not noise.

- Your crash decision selected J and O for SGD 15,500, and you explicitly rejected activity F despite it being the cheapest per day - because it is not on the critical path.
- You rejected fast-tracking the compliance review and cited both the mandatory dependency and the SGD 26,000 historical rework cost.